

# Localised Dipoles (QM)

Paul Haskins

March 14, 2025

$$\epsilon_m = -g\vec{\mu}mH \quad (1)$$

$m$  is qm number,  $H$  is constant magnetic field  $g$  is gyro magnetic ratio.

Fully localised, no interaction between particles

$$Z = \zeta^N \zeta(\beta, V, H, N = 1) \quad (2)$$

$$x = \beta g \vec{\mu}_0 J H$$

$$\zeta = \frac{\sinh\left(\frac{\frac{2J+1}{H}X}{\sinh\left(\frac{x}{2J}\right)}\right)}{\sinh\left(\frac{x}{2J}\right)} \quad (3)$$

$$\langle M \rangle = \frac{1}{\beta} \left( \frac{\partial}{\partial H} \ln \zeta \right)_{\beta, V} = g \vec{\mu} J \left[ \frac{2J+1}{2J} \coth\left(x \left( \frac{2J+1}{2J} \right)\right) - \frac{1}{2J} \coth\left(\frac{x}{2J}\right) \right] \quad (4)$$

We define the brillion function of order  $J$

$$B_J(x) = \left[ \frac{2J+1}{2J} \coth\left(x \left( \frac{2J+1}{2J} \right)\right) - \frac{1}{2J} \coth\left(\frac{x}{2J}\right) \right] \quad (5)$$

We can take limits of  $x \gg 1$  Strong magnetic vs temperature.

$$B_J(x) \approx 1, \forall J \quad (6)$$

Magnetic saturation

$x \ll 1$  weak magnetic vs temperature.

$$B_J(x) \approx \frac{1}{3} \left( \frac{J+1}{J} \right) x + \mathcal{O}(x^2) \quad (7)$$

$\Rightarrow$  keeping only the leading term

$$\langle M \rangle \approx \frac{g^2 \vec{\mu}_0^2}{3k} J(J+1) H \frac{1}{T} \quad (8)$$

No qualitative distance but only the pre-factor of  $\frac{1}{T}$  different systems have different constants.  
This is the difference between the QM and the classical Curie law.

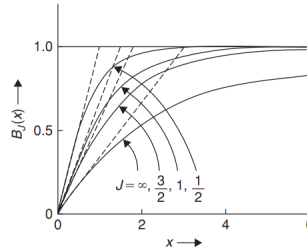


FIGURE 3.8 The Brillouin function  $B_J(x)$  for various values of  $J$ .

Figure 1: pg 75 beal

When is the classical limit obtained. When we make the spectrum almost continuous. This occurs with a large spectrum and the spectrum is very dense.

We need the following limits  $J \rightarrow \infty$  and  $g \rightarrow 0$  ( $\epsilon_m = g m \vec{\mu}_0 H$ ).

$B_J(x)$  independent of  $J$ .

$$\coth\left(\frac{x}{2J}\right) \approx \frac{2J}{x} \quad (9)$$

with  $\frac{x}{2J} \ll 1$   
In this limit we have that

$$\langle M \rangle \approx \vec{\mu} \mathcal{L}(x) \quad (10)$$

$$gJ\vec{\mu}_0 == \vec{\mu} \quad (11)$$

classical dipole moment.

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There is always a lower bound in E, but now we have an upper bound. This can lead to unusual behavior.

Fix  $J = \frac{1}{2}$ .  $\implies$  only 2 values for the energy. (fermions)

$$\epsilon_i = \begin{cases} -\vec{\mu}_0 H & == -\epsilon_0 \\ +\vec{\mu}_0 H & == \epsilon_0 \end{cases} \quad (12)$$

Thermodynamic limit - canonical ensemble

$$Z = \zeta^N = \left( \frac{\sinh 2x}{\sinh x} \right)^N = 2^N \cosh^N(x), x == \beta \vec{\mu}_0 H = \beta \epsilon_0 \quad (13)$$

$$\mathcal{F} = -kT \ln Z = -kTN \ln [2 \cosh \beta \epsilon_0] \quad (14)$$

$$S = - \left( \frac{\partial \mathcal{F}}{\partial T} \right)_{V,H,N} = NK [\ln (2 \cosh (\beta \epsilon_0)) - \beta \epsilon_0 \tanh (\beta \epsilon_0)] \quad (15)$$

$$\langle E \rangle = F + TS = - \frac{\partial \ln Z}{\partial \beta} = -N \epsilon_0 \tanh (\beta \epsilon_0) = -N \langle M \rangle H \quad (16)$$

$$\langle M \rangle = - \frac{1}{N} \left( \frac{\partial \mathcal{F}}{\partial H} \right)_{\beta,V,N} = \vec{\mu}_0 \tanh (\beta_0) \quad (17)$$

$$C_V = \left( \frac{\partial \langle E \rangle}{\partial T} \right)_{V,H,N} = \frac{Nk (\beta \epsilon_0)^2}{\cosh^2 (\beta \epsilon_0)} \quad (18)$$

thus in the limit  $\beta \epsilon_0 \ll 1$

$$S \approx Nk [\ln (2) - (\beta \epsilon_0)^2] \approx Nk \ln 2 \quad (19)$$

and if  $\beta_0 \gg 1$

$$S \approx Nk [\beta \epsilon_0 - \beta \epsilon_0] = 0 \quad (20)$$

doesn't only happens for low temp but also for strong magnetic fields.

We now examine in the micro canonical ensemble

$$S = k \ln \Omega \quad (21)$$

for the second S there would only be one microstate. We can also examine for the first S.

$$\Omega = \begin{cases} 2^N & \text{if } kT \gg \epsilon_0 \\ 1 & kT \ll \epsilon_0 \end{cases}$$

(23)

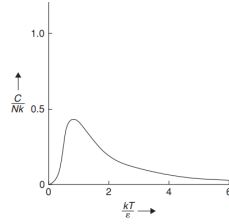


FIGURE 3.13 The specific heat of a system of magnetic dipoles (with  $J = \frac{1}{2}$ ) as a function of temperature.

Figure 2: 3.2.png

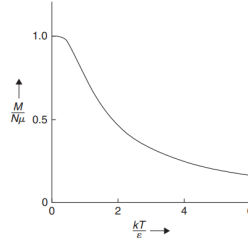


FIGURE 3.12 The magnetization of a system of magnetic dipoles (with  $J = \frac{1}{2}$ ) as a function of temperature.

Figure 3: 3.2.png

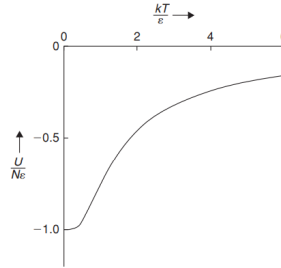


Figure 4: 3.2.png

Solve for  $T \langle E \rangle \Rightarrow$

$$\frac{1}{T} = \frac{k}{2\epsilon_0} \ln \left[ \frac{1 - \frac{\langle E \rangle}{N\epsilon_0}}{1 + \frac{\langle E \rangle}{N\epsilon_0}} \right] \quad (24)$$

\* \* \*  $T < 0$  ! Characteristic feature of systems who energy has an upper bound .

$$S = -\frac{N\epsilon_0 \langle E \rangle}{2N\epsilon_0} \ln \left[ \frac{N\epsilon_0 + \langle E \rangle}{2N\epsilon_0} \right] - \frac{N\epsilon_0 - \langle E \rangle}{2N\epsilon_0} \ln \left[ \frac{N\epsilon_0 - \langle E \rangle}{2N\epsilon_0} \right] \quad (25)$$

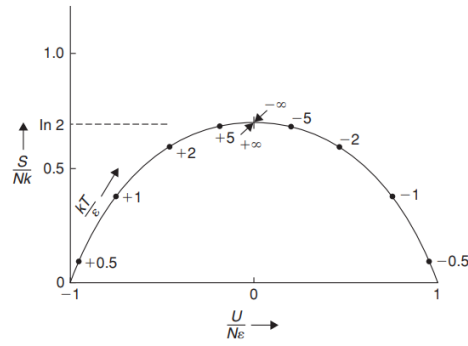


FIGURE 3.14 The entropy of a system of magnetic dipoles (with  $f = \frac{1}{2}$ ) as a function of energy. Some values of the parameter  $kT/\epsilon$  are also shown in the figure. The slope at the two endpoints diverges since both ends represent zero temperature but it is difficult to see due to the logarithmic nature of the divergence.

Figure 5: the slope of the graph is  $\frac{1}{T}$ .  $\langle E \rangle = -N\epsilon_0$ ,  $S = 0$ ,  $T = 0$ .  
 as  $\langle E \rangle$  increases,  $S$  and  $T$  increase. Max is reached as  $T \rightarrow \infty$ .  $s \approx Nk \ln 2$

But when  $\langle E \rangle \rightarrow 0^+$ , then  $\frac{\partial S}{\partial E} \propto \frac{1}{T} < 0$ .  
 Further increasing  $\langle E \rangle$   $S(E)$  decreases and  $T$  remains negative.

The RHS has hotter systems.  $T$  increase on left up to infity on left.  
 If we add a kinetic term to hamitonian then we no longer have an upper bound.